

Anurag Choudhury

+91-7008439309 | 24110387@outr.ac.in | Portfolio | GitHub | LinkedIn | Bhubaneswar, India

Education

Odisha University of Technology and Research (OUTR) Bhubaneswar, India
B.Tech in Computer Science Engineering (AI/ML) | CGPA: 9.26/10 (till 2nd Year) 2024 – 2028
 – **Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Organization & Architecture, Object-Oriented Programming, Discrete Mathematics, Data Analysis and Interpretation

Experience

iplairani.com June 2026 – Present
AI/ML Intern Remote
 – Undergoing training in **Large Language Models (LLMs), Agentic AI, and Data Science**, gaining hands-on experience with modern AI frameworks, agentic workflows, and machine learning concepts.

Projects

Cognee-MLPS_NOC | *Cognee, LangGraph, Python, Multi-Agents, Knowledge Graph*  Demo
 – Participated in the **WeMakeDevs Hackathon**, developing an **AI-powered MPLS Network Operations Copilot** that leverages persistent memory and knowledge graphs for intelligent network troubleshooting.
 – Built a production-ready memory architecture using **Cognee, LangGraph, FastAPI, and Gradio**, integrating **PGVector (Neon)** for semantic vector retrieval and **Neo4j Aura** for persistent knowledge graph storage.
 – Implemented an end-to-end AI memory lifecycle with **remember()**, **recall()**, **improve()**, and **forget()**, enabling context-aware incident analysis, long-term memory, feedback-driven learning, and memory management for network operations.

Multi Agent Research System | *LangChain, Python, AI Agents*  Demo
 – Developed and deployed a **multi-agent research platform** automating end-to-end topic exploration — from web scraping and content analysis to structured report generation.
 – Built intelligent agents for search, webpage scraping, reading, analysis, and summarization, significantly reducing manual research time for complex topics.
 – Implemented pipelines aggregating information from **multiple web sources** and producing detailed summaries with source references and clickable citations.
 – Designed an interactive frontend with real-time processing and **automated critique mechanisms** to iteratively improve research quality and output relevance.

FIFA World Cup 2026 Analytics & Prediction | *Python, Scikit-learn, Pandas, NumPy*  GitHub
 – Built an end-to-end solution for the **Zindi FIFA World Cup 2026 Prediction Challenge**, a **closed-data competition** using only historical FIFA World Cup data to predict each team's **total goals scored** and **furthest tournament stage**.
 – Engineered tournament-specific features and benchmarked **CatBoost, XGBoost, LightGBM, Random Forest, ExtraTrees**, and **HistGradientBoosting**, selecting **CatBoost Regressor (Goal MAE: 2.35/2.73/2.75)** and **CatBoost Classifier (Stage Accuracy: 50.0%/46.9%/46.9%)** based on historical World Cup validation (2014, 2018, 2022).
 – Generated the competition submission files for tournament predictions and currently hold **Rank 153** on the Zindi leaderboard, with final standings to be determined after the FIFA World Cup 2026 concludes.

TamilNadu State Elections 2026 Analysis | *Pandas, Numpy*  Analysis
 – Participated in the **Codebasics Resume Project Challenge**, building an interactive **Tamil Nadu 2026 Election analytics dashboard** to transform constituency-level election data into actionable insights.
 – Engineered a data pipeline by **scraping, cleaning, and integrating** multiple election datasets, resolving constituency mismatches and reconstructing voter turnout statistics for all 234 constituencies.
 – Performed comparative analysis between the 2021 and 2026 elections, uncovering key trends including **163 constituency flips**, increased voter participation, reduced victory margins, and changing vote-share patterns.

End-to-End Data Warehouse & Analytics Solution | *Python, SQL, Medallion Architecture*

 [Github](#)

- Designed and implemented a **multi-layer data warehouse** using Medallion Architecture (Bronze, Silver, Gold), enabling structured reporting across 3 distinct data layers.
- Engineered automated data ingestion and transformation pipelines, consolidating raw-to-analytics workflows into a single modular pipeline.
- Applied data engineering best practices (partitioning, indexing, schema design) to generate actionable business insights from processed datasets.

SQL Advanced Data Analytics Project | *SQL*

 [Dashboard](#)

- Built an end-to-end SQL analytics solution on a Medallion Architecture data warehouse, converting curated Gold-layer data into business-ready analytical reports.
- Implemented advanced SQL techniques including **window functions, CTEs, subqueries, ranking, and aggregations** to derive insights across sales, customers, and products.
- Developed a **dashboard** for interactive visualization of KPIs, trend analysis, and executive reporting to support business decision-making.

Technical Skills

Languages: Python, C++, SQL

Generative AI: LLMs, LangChain, LangGraph

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Plotly

Databases: MySQL, PostgreSQL, OracleSQL

Tools & Platforms: Git, GitHub, Flask

Core Concepts: DSA, OOP, DBMS, OS, Computer Architecture, Data Analysis

Others: Finance, Stock Market Fundamentals

Certifications

Software Development Fundamentals

Certificate

HackerRank SQL — Basic, Intermediate & Advanced

Basic | Intermediate | Advanced

Achievements & Activities

- **SIH Internal Hackathon 2025** — Selected in the Smart India Hackathon internal round, competing among 100+ teams college-wide.
- **Hackyard 2026** — Ranked in **Top 10 teams** out of all participating teams in the inter-college hackathon.
- **GirlScript Summer of Code (GSSoC) 2026** — Selected as an **open-source contributor**